

$$\begin{aligned}
 & \frac{v^2}{\alpha} \leq t \\
 & \leq e^{\frac{t}{2\alpha} - \frac{t}{\alpha}} \\
 & = e^{-\frac{t}{2\alpha}} \\
 & P(X - \mu \geq t) \leq \begin{cases} e^{-\frac{t^2}{2v^2}} & t \leq \frac{v^2}{\alpha} \\ e^{-\frac{t}{2\alpha}} & t > \frac{v^2}{\alpha} \end{cases}
 \end{aligned}$$

Exercise 2.3.

$$E(X) = \dots \quad X \geq 0$$

$$E(e^{\lambda(X - \mu)}) < \infty$$

$$\lambda \in [-\beta, \beta]$$

$$\inf_{k=0,1,\dots} \frac{E\{|X|^k\}}{s^k} \leq \inf_{\lambda > 0} \frac{E\{e^{\lambda X}\}}{e^{\lambda s}}$$

Proof:

$$\begin{aligned}
 e^{\lambda X} &= \sum_{k=0}^{\infty} \frac{\lambda^k X^k}{k!} \Rightarrow E(e^{\lambda X}) = E\left(\sum_{k=0}^{\infty} \frac{\lambda^k X^k}{k!}\right) \\
 &= \sum_{k=0}^{\infty} E(|X|^k) \frac{\lambda^k}{k!} \geq \sum_{k=0}^{\infty} r s^k \frac{\lambda^k}{k!} = \underbrace{r e^{s\lambda}}_{\text{Series}}
 \end{aligned}$$

$\swarrow$ Abs. Conv. $\quad \downarrow$ $X \geq 0$

$$\sum_{k=0}^{\infty} a_k \text{ is called abs. conv.} \Leftrightarrow \sum_{k=0}^{\infty} |a_k| \text{ exists.}$$

$$x \in (0, 1]$$

$$\inf_{x \in (0, 1]} x = 0$$

$$r := \inf_{k=0,1,\dots} \frac{\mathbb{E}\{|X|^k\}}{\delta^k} \quad \forall k \in \{0,1,\dots\} : \frac{\mathbb{E}\{|X|^k\}}{\delta^k} \geq r.$$

$$\sum_{k=0}^{\infty} \frac{\delta^k \lambda^k}{k!} = e^{\lambda \delta}$$

$$\Rightarrow \forall \lambda \quad \frac{\mathbb{E}\{e^{\lambda X}\}}{e^{\lambda \delta}} \geq \underbrace{r}_{\inf_{k=0,1,\dots}} \frac{\mathbb{E}\{|X|^k\}}{\delta^k}$$

$$\Rightarrow \inf_{\lambda > 0} \frac{\mathbb{E}\{e^{\lambda X}\}}{e^{\lambda \delta}} \geq \inf_{k=0,1,\dots} \frac{\mathbb{E}\{|X|^k\}}{\delta^k} \quad \square$$

Motivation :

- Computing the moment gen. func. $\wedge$
upper bounding it is not always easy.

Assumption $\exists b \quad \forall k \in \{2,3,\dots\} \quad \left| \mathbb{E}\{(X-\mu)^k\} \right| \leq \frac{1}{2} k! \sigma^2 b^{k-2}$

Example: if X is bounded, i.e. $|X-\mu| \leq b$
this assumption is satisfied.

$$\begin{aligned} \left| \mathbb{E}\{(X-\mu)^k\} \right| &= \left| \int (x-\mu)^k (x-\mu)^{k-2} p_X(x) dx \right| \\ &\leq \int (x-\mu)^2 \underbrace{|x-\mu|^{k-2}}_{\leq b^{k-2}} p_X(x) dx \end{aligned}$$

$$\leq b^{k-2} \int (x-\mu)^2 p_X(x) dx = b^{k-2} \sigma^2$$

$$\leq \frac{k!}{2} \sigma^2 b^{k-2}$$

$$k \in \{2, \dots\}$$

□

$$\mathbb{E}\{e^{\lambda(X-\mu)}\} = \mathbb{E}\left\{\sum_{k=0}^{\infty} \frac{\lambda^k (X-\mu)^k}{k!}\right\}$$

$$= 1 + \frac{\lambda^2 \sigma^2}{2} + \sum_{k=3}^{\infty} \frac{\mathbb{E}(X-\mu)^k \lambda^k}{k!}$$

$$\leq 1 + \frac{\lambda^2 \sigma^2}{2} + \sum_{k=3}^{\infty} \frac{|\mathbb{E}(X-\mu)^k| |\lambda|^k}{k!}$$

$$\leq 1 + \frac{\lambda^2 \sigma^2}{2} + \sum_{k=3}^{\infty} \frac{\cancel{1/2} \cancel{k!} \sigma^2 b^{k-2} |\lambda|^k}{\cancel{k!}}$$

$$= 1 + \frac{\lambda^2 \sigma^2}{2} + \frac{\lambda^2 \sigma^2}{2} \sum_{k=3}^{\infty} |\lambda|^{k-2} b^{k-2}$$

$$= 1 + \frac{\lambda^2 \sigma^2}{2} \left[\underbrace{\sum_{k=0}^{\infty} |\lambda|^k b^k}_1 \right]$$

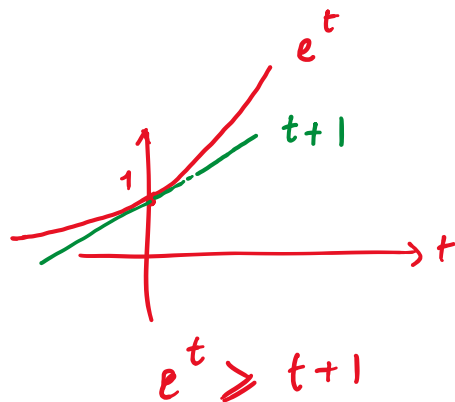
$$\frac{1}{1 - |\lambda|b}$$

$$|\lambda|b < 1$$

$$|\lambda| < 1/b$$

$$= 1 + \frac{\lambda^2 \sigma^2 / 2}{\underbrace{1 - |\lambda|b}_t}$$

$$\leq e^{\frac{\lambda^2 \sigma^2 / 2}{1 - |\lambda|b}}$$



$$|\lambda| < 1/b$$

$$\leq e^{\frac{2\lambda^2\sigma^2}{2}} = e^{\frac{\lambda^2(\sqrt{2}\sigma)^2}{\sigma'^2/2}} \quad |\lambda| < \frac{1}{2}b$$

$$|\lambda|b < \frac{1}{2}$$

$$1 - |\lambda|b > \frac{1}{2}$$

$$\Rightarrow X \sim \text{subE}(\sqrt{2}\sigma, 2b)$$

Theorem: Suppose that $\exists b \forall k \in \{2, \dots\}$ Bernstein

$$|\mathbb{E}(X-\mu)^k| \leq \frac{1}{2} k! \sigma^2 b^{k-2}.$$

$$\text{Then } \forall t > 0, \mathbb{P}(|X-\mu| \geq t) \leq 2 e^{-\frac{t^2}{2(\sigma^2 + bt)}}.$$

Proof: $\mathbb{P}(X-\mu \geq t) \leq \mathbb{E}(e^{\lambda(X-\mu)}) \cdot e^{-\lambda t}$

$$\leq e^{\frac{\lambda^2\sigma^2/2}{1-|\lambda|b}} \cdot e^{-\lambda t}$$

$$\lambda^* = \frac{t}{bt + \sigma^2} = e^{\frac{\frac{1}{2}t^2}{(bt + \sigma^2)^2 \cancel{\sigma^2}} / \left(1 - \frac{bt}{bt + \sigma^2}\right)}.$$

$$e^{-\frac{t^2}{bt + \sigma^2}} \quad \frac{\cancel{\sigma^2}}{bt + \sigma^2}$$

$$= e^{-\frac{t^2}{2(bt + \sigma^2)}} \quad \square.$$

Johnson - Lindenstrauss Lemma

$$u^1, \dots, u^n \in \mathbb{R}^d$$

$$\forall u^i, u^j \quad 1 - \delta \leq \frac{\|F(u^i) - F(u^j)\|}{\|u^i - u^j\|} \leq 1 + \delta$$

$$X \sim \chi_n^2$$

$$\mathbb{P}(|X - \mu| \geq t)$$

$$\left. \begin{array}{l} X_1 = Z_1^2 \\ \vdots \\ X_n = Z_n^2 \end{array} \quad \begin{array}{l} Z_1 \sim \mathcal{N}(0,1) \\ \vdots \\ Z_n \sim \mathcal{N}(0,1) \end{array} \right\} \text{iid}$$

$$X = \sum_{i=1}^n Z_i^2 \sim \chi_n^2 \quad \mathbb{E}\left\{\frac{1}{n}X\right\}$$

$$\mathbb{P}\left\{\left(\frac{1}{n} \sum Z_i^2\right) - 1 \geq t\right\}$$

$$Z_1^2 \sim \text{sub E}(2, 4)$$

$$Z_1^2 + \dots + Z_n^2 \sim \text{sub E}(2\sqrt{n}, 4)$$

$$\mathbb{P}\left(\frac{Z_1^2 + \dots + Z_n^2}{n} - \frac{n}{n} \geq \frac{t}{\frac{n}{t'}}\right) \leq e^{-\frac{t^2}{2(4n)}} = e^{-\frac{t^2}{8n}} \quad t < t'$$

$t = t'n$ $= e^{-\frac{t'^2 n}{8}}$

$$A_{m \times d} \quad A_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0,1)$$

$$F(u) = \frac{1}{\sqrt{m}} A u$$

$$1 - \delta \leq \frac{\|F(u^i) - F(u^j)\|_2^2}{\|u^i - u^j\|_2^2} \leq 1 + \delta \quad \text{w.h.p}$$